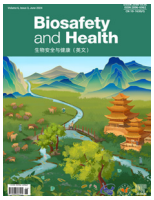
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Original Research

Beyond keywords: Semantic classification with large language models for mapping synthetic biology literature

Wenbing Ma [a], Pengyuan Nie [b], Rongling Yin [a], Lijuan Li [a,*], Lei Wang [a,*]

[a] Academy of Military Medical Sciences, Beijing 100850, China
[b] The Xiamen University, School of Film, Xiamen 361000, China

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ABSTRACT

Emerging interdisciplinary fields such as synthetic biology pose a fundamental challenge for bibliometric analysis: their boundaries are inherently blurred, and a substantial portion of their relevant literature cannot be captured by conventional keyword-based retrieval. To address this problem, we developed a two-step pipeline that combines a hybrid greedy-frequency keyword selection algorithm for high-recall candidate retrieval with a fine-tuned large language model (LLM) for semantic-level classification. On a curated test set, the fine-tuned LLM achieved a macro-F1 of 0.97, outperforming traditional classifiers. In a real-world validation experiment using the complete publication set of *ACS Synthetic Biology* as a fully positive corpus, the pipeline recovered 57.81% of the journal's articles — 2.1 times more than a single keyword search (27.20%) and 1.8 times more than the best previously published keyword set (32.83%) — while maintaining a false positive rate of only 0.76% on a large scale borderline negative corpus of 475,019 articles. Applying the pipeline to an OpenAlex snapshot yielded a curated corpus of synthetic biology literature, from which we constructed a bibliometric landscape map encompassing annual publication trends, international collaboration networks, a keyword co-occurrence network, a co-citation network, and a thematic strategic diagram. The proposed framework is modular, quantitatively validated, and potentially transferable to other interdisciplinary life science domains such as proteomics, brain science, immunoengineering, and nanobiotechnology.

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1. Introduction

Synthetic biology is a rapidly evolving interdisciplinary field that applies engineering principles to biology in order to design and construct new biological parts, devices, and systems, or to rewire existing ones for useful purposes [1]. It merges molecular biology, genetics, systems biology, and computer science, with the ultimate goal of making biology easier to engineer in a predictable and scalable way. At its core, synthetic biology shifts biology from a descriptive science to a constructive one.

Synthetic biology has drawn sustained policy attention for both its economic promise [2–4] and its biosafety risks [5,6]. The economic potential of synthetic biology has been codified in national strategies [7], including the U.S. *Executive Order 14081* (2022) [8], China's *14th Five Year Plan for Bioeconomy Development* [9,10], and the European Commission's *2024* Communication on boosting biotechnology and biomanufacturing [11].

However, the same technological advances that drive the bioeconomy also heighten biosafety concerns, compelling governments to strengthen oversight [12–15]. China's *Biosafety/Biosecurity Law* (2020) creates a comprehensive oversight framework covering pathogen laboratories, invasive species, and biotechnological research activities, signaling a coordinated national response [16]. In the United States, the National Academies of Sciences, Engineering, and Medicine warned in *Biodefense in the Age of Synthetic Biology (2018)* that falling DNA synthesis costs and accessible genome-editing tools lower the barriers to creating pandemic-capable viruses [17]. The U.S. Department of Health and Human Services (HHS) issued the updated *Screening Framework Guidance for Providers and Users of Synthetic Nucleic Acids* in 2023 [18], which established baseline standards for the gene synthesis industry. This was followed in 2024 by the White House Office of Science and Technology Policy (OSTP) *Framework for Nucleic Acid Synthesis Screening* [19]. Meanwhile, the *Global Guidance Framework for the Responsible Use of the Life Sciences (2022)*, issued by the World Health Organization (WHO), calls for biorisk management that keeps pace with synthetic biology's advance [20]. These measures collectively illustrate that policy is now trying to build safety into the bio-revolution from the start—precisely because the failure to do so would

* Corresponding authors: Academy of Military Medical Sciences, Beijing 100850, China (L. Wang and L. Li).
E-mail addresses: juanlah@aliyun.com (L. Li), wangleienjoy@163.com (L. Wang).

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undermine both public trust and the very bioeconomy governments seek to grow.

Bibliometric analysis provides a quantitative means to map the scientific literature and inform science policy [21,22]. Bibliometric mapping of scientific articles based on keywords and technical terms in titles and abstracts is now frequently used to chart scientific fields [23]. However, emerging interdisciplinary fields like synthetic biology are inherently difficult to define with precision due to their blurred boundaries, which makes mapping them extremely challenging [24,25]. Traditional keyword-based methods are scalable but often fail to capture semantically relevant work that does not carry explicit domain-specific terms. For instance, different authors may use different keywords to describe conceptually similar work [26–29], and more critically, a substantial body of technically relevant research does not explicitly carry the expected domain-specific keywords in its title, abstract, or indexed terms. Shapira et al. [24] developed a well-designed approach using a core set of papers to benchmark and enumerate a collection of keyword combinations; however, there is no quantitative method to prove that these keywords are comprehensive enough to cover most papers in the synthetic biology field, which hinders the bibliometric process. Expert-involved approaches that combine lexical queries with citation–based extension and expert-validated clustering can improve precision, but they remain labour–intensive, difficult to reproduce at scale, and may still be subject to expert disagreement on borderline cases [30]. The use of field-dedicated journals as proxies for domain boundaries offers simplicity but systematically misses relevant work published in generalist or neighbouring journals. Large language models (LLMs) have brought significant advances to natural language processing and have been widely adopted for text classification tasks [31,32], achieving impressive performance even in specialized scientific domains [33]. Their potential application to bibliometrics, however, remains at an exploratory stage. A few recent studies have begun to experiment with LLMs as virtual domain experts for tasks such as research quality assessment, thematic discovery and semantic classification of research content [34–40]. While these efforts demonstrate the feasibility of leveraging LLMs for bibliometric purposes, they also reveal a fundamental tension: LLMs may be strong semantic interpreters, but they are not naturally suited to serve as systematic retrievers. Bibliometric analysis is fundamentally a statistician's task—it requires exhaustive, reproducible, and unbiased coverage of a defined literature corpus. This leads to the two-step pipeline, combining broad keyword retrieval with semantic classification.

In this paper, we aim to develop a bibliometric pipeline that can map synthetic biology literature with both high coverage and high precision. Specifically, we propose a two step strategy: a hybrid greedy frequency algorithm first constructs a quantitatively validated seed keyword set to retrieve a broad candidate pool; a fine tuned LLM classifier then semantically filters this pool to exclude irrelevant articles. This method can quantitatively balance comprehensiveness and accuracy. Taking synthetic biology as a case study, we demonstrate the pipeline's practical utility through a comprehensive bibliometric landscape analysis. While this study focuses on synthetic biology, the modular nature of the pipeline suggests its broader applicability to other interdisciplinary fields that resist simple keyword based delineation.

2. Materials and methods

2.1. Overview of the two step pipeline

The proposed pipeline addresses the dual challenge of field delineation — comprehensive coverage and high precision — through a two stage architecture (Fig.1). The first stage is designed for high recall candidate retrieval: a seed keyword set, constructed through a hybrid greedy frequency algorithm, queries large scale bibliographic databases to capture a broad pool of potentially relevant articles with minimal omissions. The second stage performs high precision semantic filtering: a fine tuned LLM, trained on a deliberately curated dataset that includes positive, borderline, and background samples, classifies each candidate article by evaluating its full semantic content against the definition of the target field. This division of labor — broad recall followed by fine grained discrimination — ensures that the resulting corpus is both representative of the domain and sufficiently pure for downstream bibliometric analysis.

2.2. Dataset construction: Positive, borderline, and background samples

To train and evaluate the article classifiers, we constructed a labeled dataset composed of three distinct sample categories that reflect the real world challenges of distinguishing synthetic biology from its closely related disciplines. Positive samples were sourced from MEDLINE by retrieving all available articles explicitly annotated with the Medical Subject Headings (MeSH) term "Synthetic Biology" (2009–2025). Borderline samples were collected from articles annotated with the parent MeSH terms "Biology" and "Bioengineering" — these terms sit above synthetic biology in the MeSH hierarchy and capture thematically adjacent but not strictly synthetic biology content. Articles already labeled as "synthetic biology" were excluded from the borderline pools to prevent label leakage. Background samples were obtained from the MeSHup dataset [41], which provides compound MeSH term annotations covering a wide range of biology- and bioengineering related subfields, thereby representing a broader negative class. After deduplication and the removal of records with missing abstracts or titles, we retained a total of 5,201 positive articles. To create a balanced yet challenging training pool, we randomly sampled an equal number of negative articles (5,201) from each of the three negative sources — Biology, Bioengineering, and Background — resulting in a 1:1:1:1 ratio across the four categories and a total pool of 20,804 articles (Table S1). This pool was then split into training and test sets using a stratified 9:1 split, preserving the label proportions (Table S2).

For the construction and coverage evaluation of the seed keyword set, we retrieved 5,697 articles indexed with the MeSH term "Synthetic Biology" from PubMed (query: "synthetic biology"[MeSH Terms]; search date: 15 February 2026; records up to 2025). MeSH terms and author supplied keywords were extracted from each record to build a keyword-article occurrence matrix, which was subsequently used in the frequency based, greedy, and hybrid keyword selection experiments.

For the real world recall validation, we used the complete set of articles published in *ACS Synthetic Biology* and indexed in the Web of Science (WoS) Core Collection as of 11 October 2025. As these articles have undergone rigorous peer review and represent the consensus of experts actively working in the field, they provide an independent knowledge channel that is distinct from MeSH based indexing. This corpus therefore serves as a fully positive set against which the coverage of different retrieval strategies can be measured. For the false positive validation, constructing a truly representative real world negative set is inherently difficult, as it would require exhaustive labeling of all articles that are not synthetic biology. We therefore focused on the model's ability to discriminate against challenging borderline cases. Specifically, we retrieved all articles indexed in Web of Science with the parent MeSH terms "Biology" and "Bioengineering." After excluding articles that were already included in our training or test sets, as well as those explicitly tagged with "Synthetic Biology," the remaining pool comprised 475,019 articles. This expansive negative space is approximately 33 times larger than the negative portion of our training set (14,042 articles), providing a stringent large scale test of the model's ability to avoid false positives among closely related but non synthetic biology content.

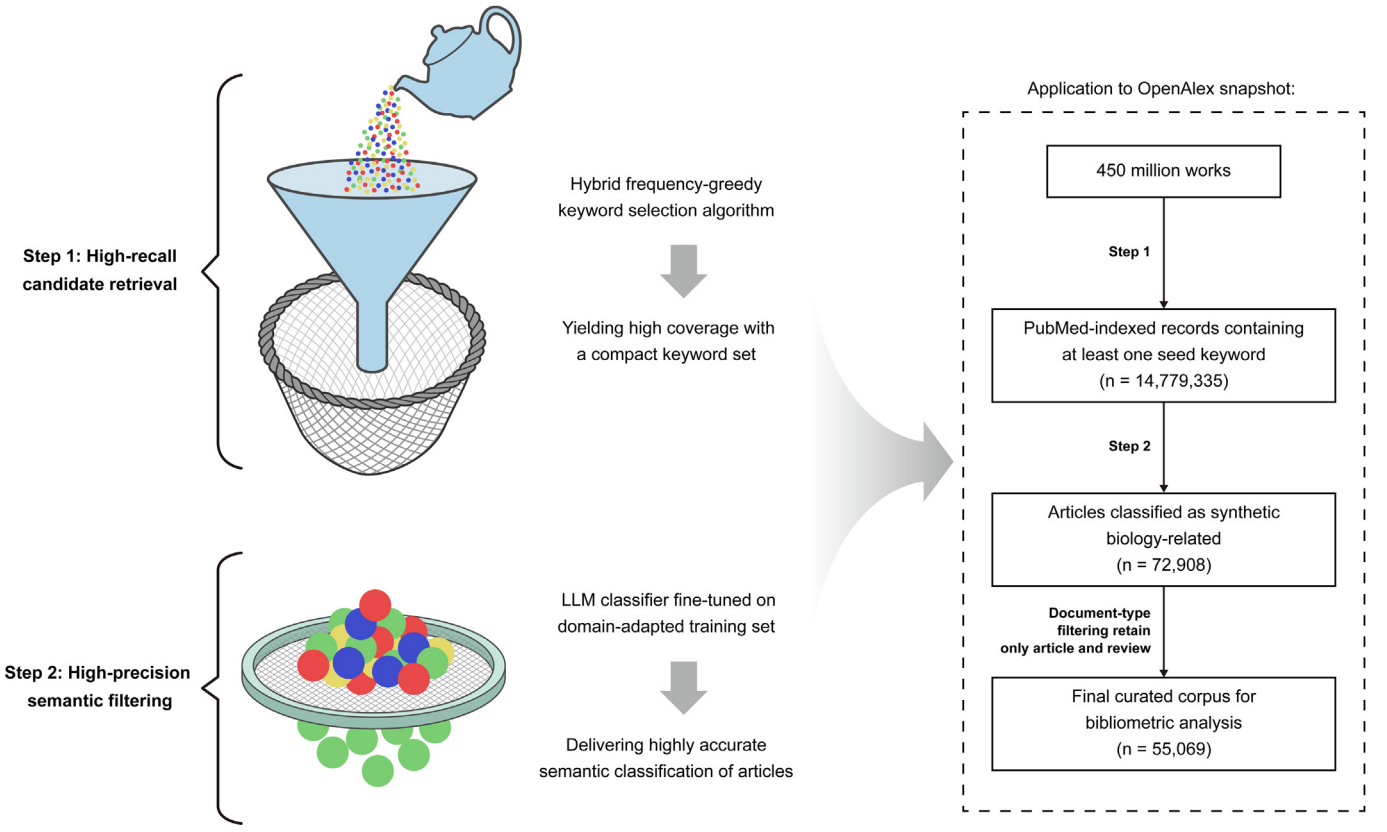


Fig. 1. Schematic overview of the two step bibliometric pipeline. Left: conceptual illustration of the two-stage architecture. Right: flow diagram showing the number of records at each stage when the pipeline was applied to the OpenAlex snapshot. Abbreviation: LLM, large language model.

For the bibliometric landscape analysis, we used a snapshot of the OpenAlex database — an open scholarly index of over 450 million works — downloaded on 26 March 2026. This snapshot was processed through our two step pipeline to retrieve synthetic biology articles, which then formed the corpus for all subsequent bibliometric analyses. During this retrieval, the first stage keyword query was restricted to PubMed indexed records to align with the domain scope of the training data and to reduce the risk of noise from out of distribution samples.

2.3. Selection of the seed keyword set

To construct a seed keyword set that can retrieve synthetic biology articles with high recall, we first built a keyword-article occurrence matrix from the MeSH-indexed records described in Section 2.2. From each record, we extracted all MeSH terms and author-supplied keywords; after normalisation (lowercasing and punctuation removal), each unique term was treated as a candidate keyword. The matrix M was defined such that $M_{i,j}=1$ if keyword i appears in article j, and 0 otherwise. From this matrix, we evaluated two baseline selection strategies and one proposed hybrid strategy (Fig. 2).

The frequency based method pre ranks all candidate keywords by their total occurrence frequency across the entire corpus and iteratively selects them in descending order; articles containing the next highest ranked keyword are marked as covered and removed from the pool.

The second, a greedy algorithm, instead selects at each step the keyword that covers the largest number of currently uncovered articles. Formally, let U be the set of articles not yet covered. The keyword k_t chosen at step t is defined as Equation (1):

$$k_t = \underset{k}{argmax} \left|\{d \in U : k \in \text{keywords}(d)\}\right| \quad (1)$$

The covered articles are then removed from U, and the procedure continues until the target coverage is achieved.

The hybrid method restricts the greedy algorithm's search space to a pre-filtered set of high-frequency keywords. Specifically, we rank all candidate keywords by total frequency and retain only the top N (e.g., the top 1,000); greedy selection is then performed on this restricted set. This constraint prevents the inclusion of extremely short or rare terms while preserving the coverage efficiency of the greedy strategy, yielding a compact seed keyword set that balances recall and noise.

To assess the stability, the full article set was randomly partitioned into three independent, non-overlapping subsets of equal size. The hybrid selection procedure was applied to each subset separately, and the resulting keyword sets were compared using two metrics: (1) pairwise Jaccard similarity indices between the sets, and (2) the consistency of individual keywords' ranks and appearance counts across splits.

2.4. LLM based article classification

Following recent work that leverages LLMs for semantic classification of research orientations [42], we frame field delineation as a binary semantic classification task. The article classifier was built upon the Qwen3 4B LLM [43], which was prompted with a unified template that jointly encodes the domain definition of synthetic biology, a binary classification rule (1 = relevant, 0 = irrelevant), and an output format restriction (response must be a single digit). Both the base Qwen3 4B and its fine tuned variant received exactly the same prompt to ensure comparability. For fine tuning, we applied low rank

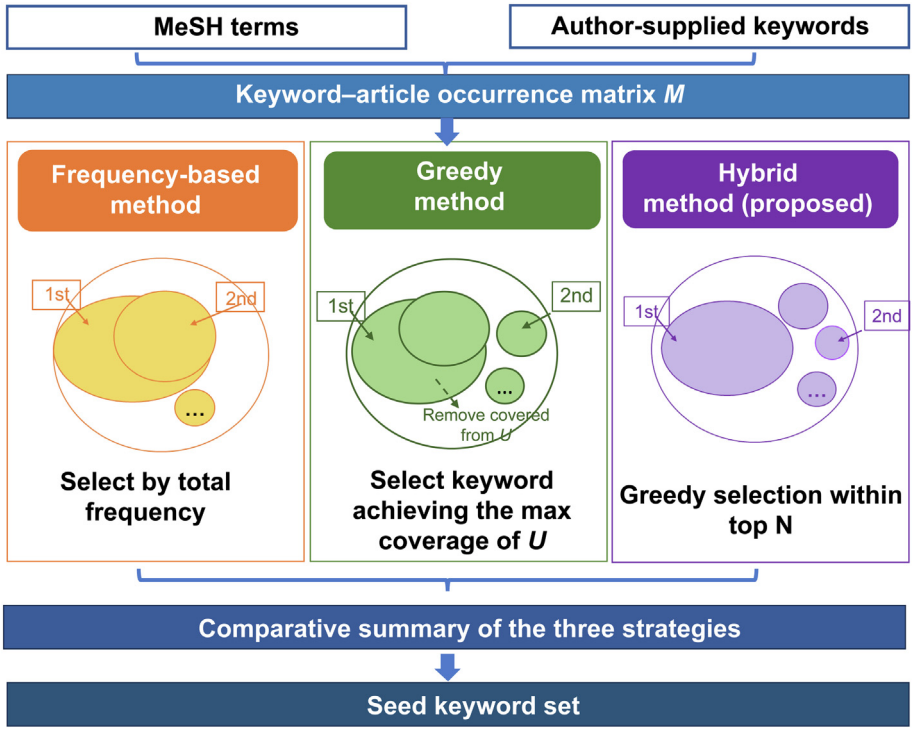


Fig. 2. Schematic illustration of the three keyword selection strategies. Abbreviation: MeSH, Medical Subject Headings

adaptation (LoRA) [44] to the pre trained Qwen3 4B using the labeled training set described above; key hyper parameters (learning rate, epochs, LoRA rank) were selected via preliminary experiments on a held out validation split. To assess whether simply increasing model scale improves classification quality, we also evaluated three larger base variants — Qwen3 8B, Qwen3 14B, and Qwen3 32B — under the identical prompt. As competitive baselines, we implemented two traditional bag of words classifiers, Complement Naïve Bayes (ComplementNB) and Multinomial Naïve Bayes (MultinomialNB), both trained on term frequency–inverse document frequency (TF IDF) features with 1 to 3 gram tokenisation.

All models were evaluated on the fixed test set. We report macro-averaged precision, recall, and F1-score, which weight both classes equally. Precision and recall are defined in Equations (2) and (3), respectively, as:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. The F1-score is the harmonic mean of precision and recall, as defined in Equation (4):

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

2.5. Bibliometric analysis framework

To characterize the synthetic biology landscape, we applied a standard bibliometric workflow to the curated corpus. All analyses were performed using the R package bibliometrix (v5.3) [45]. We first examined the annual scientific production to quantify the growth trajectory of the field. A country collaboration network was constructed based on co–authorship at the country level. The conceptual structure of the field was explored through a keyword co-occurrence network. The intellectual structure was further investigated via a co-citation network of the 50 most frequently co-cited references, with Walktrap community detection [46]. Thematic focus and maturity of different research areas were assessed using a thematic map based on author–supplied keywords, with Louvain clustering applied to identify themes, which were then plotted according to their centrality and density in a strategic diagram. For the co-citation network, community labels were assigned by the authors based on manual inspection of the titles and abstracts of the clustered references, representing a domain–knowl edge–guided interpretation of the algorithmic groupings.

3. Results

3.1. Performance of the LLM-based article classifier

We compared the performance of four classifiers evaluated on the same test set of 2,081 articles, of which 520 (25%) were positive samples (Table 1). The base Qwen3 4B model was prompted with a specially designed context (Fig. 3) that combines a domain definition of synthetic biology with a binary classification rule and a strict output format restriction. Three larger variants — Qwen3 8B, Qwen3-14B, and Qwen3-32B — were evaluated under the identical prompt to examine whether increased model scale alone improves classification quality. All four base models yielded outputs that were uniformly well structured, producing only the labels “0” or “1” as requested. Moving from the 4B to the 14B variant brought a modest but consistent improvement in macro averaged F1 (from 0.86 to 0.89); however, the performance of the 32B model (F1 = 0.89) was virtually identical to that of the 14B model, suggesting that the additional capacity of the larger model contributes little beyond what can already be achieved by the mid sized variant under this prompt only setting. This result indicates that the prompt design itself is effective, but that simply scaling up the base model without task-specific fine-tuning yields diminishing returns.

Fine tuning the Qwen3 4B model on our labeled training data using the LoRA pipeline substantially improved classification quality, raising

Table 1
Performance of the LLM-based article classifier.

Model	Precision	Recall	F1	Pos. size/All
Qwen3-4B	0.85	0.90	0.86	520/2,081
Qwen3-8B	0.86	0.91	0.88	520/2,081
Qwen3-14B	0.87	0.92	0.89	520/2,081
Qwen3-32B	0.86	0.92	0.89	520/2,081
ComplementNB (TF-IDF, 1-3 gram)	0.90	0.95	0.92	520/2,081
MultinomialNB (TF-IDF, 1-3 gram)	0.91	0.95	0.93	520/2,081
LLM fine-tuned (Qwen3-4B)*	**0.97**	**0.96**	**0.97**	520/2,081

Note: * indicates our proposed classifier; bold font indicates the best of all results. Abbreviations: Pos., positive; LLM, large language model; TF-IDF, term frequency–inverse document frequency; ComplementNB, complement naïve Bayes; MultinomialNB, multinomial naïve Bayes.

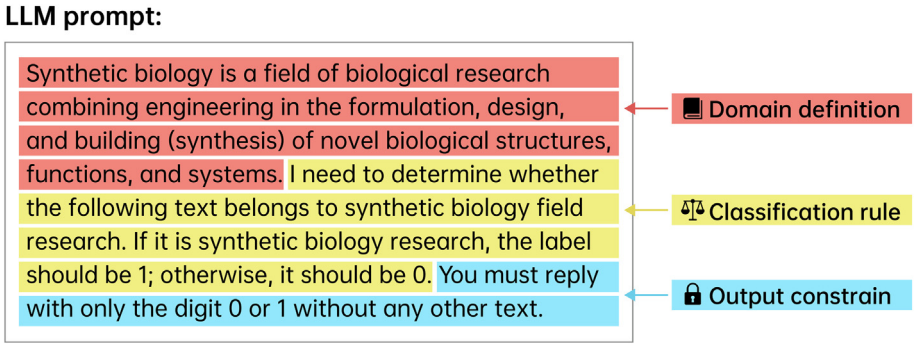


Fig. 3. Unified prompt design for both base and fine-tuned large language models (LLMs)

macro averaged precision to 0.97, recall to 0.96, and F1 score to 0.97 (Table 1). This fine tuned 4B variant outperformed the two traditional bag of words classifiers — ComplementNB (F1 = 0.92) and MultinomialNB (F1 = 0.93) — as well as all the untuned larger Qwen3 models. Notably, the fine tuned LLM achieved the highest precision among all models while maintaining a recall comparable to the best Naïve Bayes classifier. This indicates that it effectively minimises false positives without sacrificing coverage of the minority class — a property that is especially desirable in automated literature retrieval pipelines, where high precision ensures that the resulting document collection is of high purity and can be directly used for downstream bibliometric tasks without manual correction. The strong performance of the fine - tuned model, in contrast to the marginal gains observed from simply increasing model size, demonstrates that domain specific fine tuning is the critical factor for adapting large pre trained language models to specialized classification tasks.

Structure of the prompt used for classifying synthetic biology articles. The prompt integrates a domain definition, a binary classification rule, and an output format restriction. Both the base Qwen3 series model (4B, 8B, 14B, and 32B) and its fine tuned variant receive exactly the same prompt template for training and inference, respectively, which guarantees comparability in the output format.

To assess whether the fine-tuned classifier genuinely navigates the blurred boundaries of synthetic biology — rather than simply distinguishing it from distantly unrelated fields — we evaluated its performance on six MeSH-indexed neighboring fields conceptually adjacent to synthetic biology: metabolic engineering, systems biology, genetic engineering, protein engineering, computational biology, and cell engineering. After removing articles co-tagged with “Synthetic Biology”, we applied both the base Qwen3-4B model and the fine-tuned conservative variant to these corpora (Table S3).

The base model massively over-predicted synthetic biology in these adjacent domains, with false positive rates ranging from 32.42% (systems biology) to 95.81% (metabolic engineering). This high base-model error rate itself reflects the substantial conceptual overlap between these neighboring fields and synthetic biology, confirming that they represent genuine “hard boundary” cases. In contrast, fine-tuning suppressed these rates to below 17% across all six fields, with particularly strong performance in computational biology (1.02%) and systems biology (3.69%). This demonstrates that the classifier has learned to effectively discriminate synthetic biology from closely related disciplines at their blurred boundaries.

3.2. Construction and coverage validation of the seed keyword set

The second question we investigate is whether a set of keywords can effectively cover the articles within the synthetic biology domain. This question is motivated by our two stage retrieval strategy: in the first stage, a curated keyword set is used to retrieve a broad pool of candidate articles with high recall; in the second stage, a fine tuned LLM acts as a precise classifier to filter out irrelevant items. Since the LLM can only process articles that have been retrieved, the keyword set must achieve near complete coverage of the target domain to avoid missing relevant literature. At the same time, the keyword set should not be so noisy as to overwhelm the downstream classifier. Designing such a keyword set is non-trivial, because overly broad terms introduce excessive noise, while overly specific terms may miss a substantial fraction of the synthetic biology research. Understanding how different keyword selection strategies balance coverage and noise is therefore essential for constructing an effective retrieval pipeline.

To address this, we evaluated three selection strategies. The first is a frequency-based approach: we iteratively select the most frequent keyword from the remaining pool of articles, adding it to the set and updating the cumulative coverage. The second is a greedy algorithm: at each step, we choose the keyword that covers the greatest number of yet–uncovered articles, rather than simply the most frequent one. We found that the comparison is robust, as the greedy based method only needs 13 keywords for 90% coverage, while the frequency based method needs 173 keywords (Fig. 4A, Table S4). However, a notable limitation of the greedy algorithm is its tendency to select excessively granular keywords. For instance, terms such as “pla,” “ai,” and “ph” appear among the top-ranked selections (Table S5, Table S6). Their

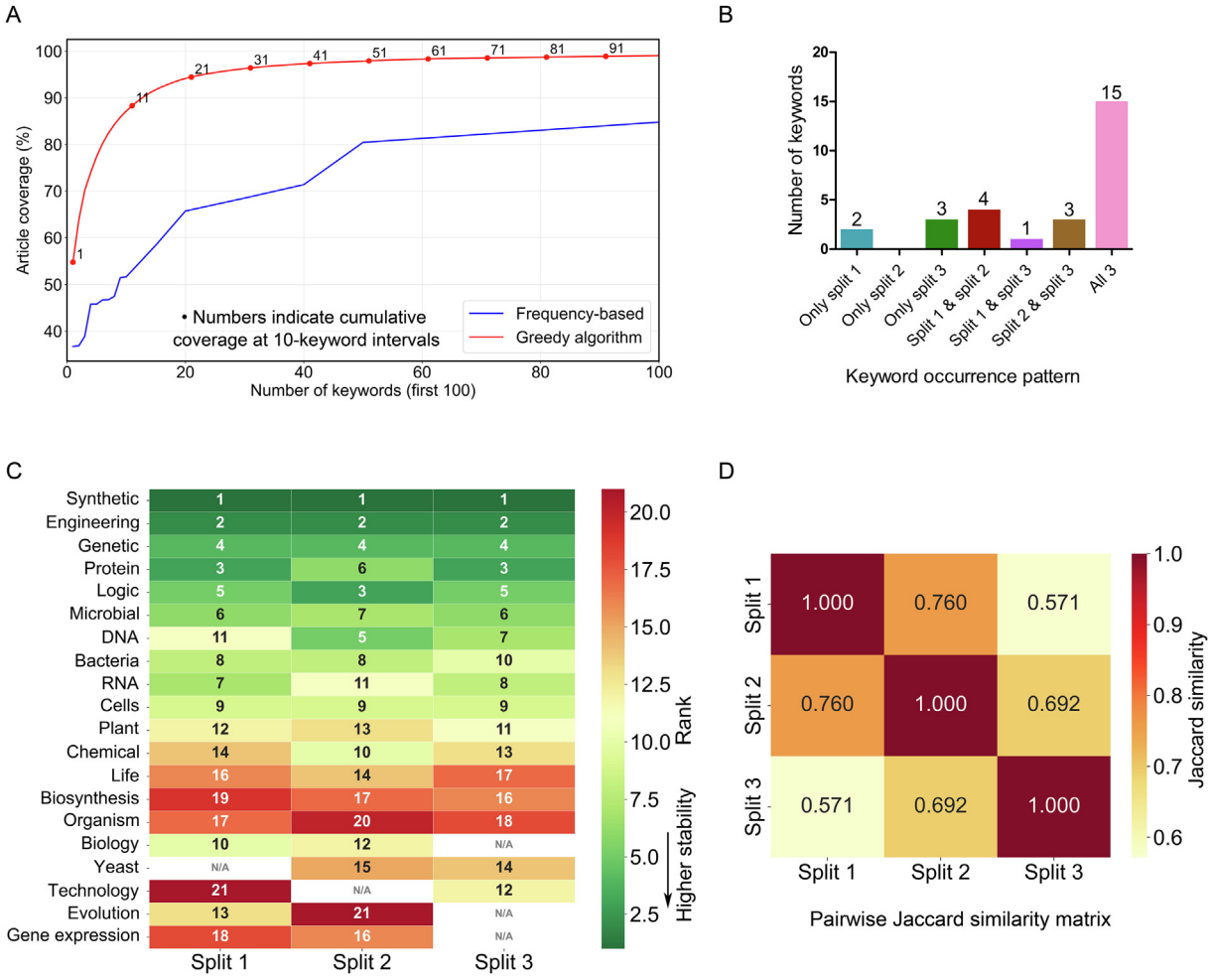


Fig. 4. Construction and coverage validation of the seed keyword set. A) Detailed comparison of keyword coverage (first 100 selected keywords). Comparison of cumulative article coverage for the two keyword selection strategies over the first 100 keywords. The greedy algorithm (red) achieves a steeper coverage increase than the frequency-based method (blue), demonstrating its ability to capture a larger fraction of the literature with fewer terms. B) Overlap analysis of the top 22 keywords selected by the hybrid algorithm across three independent data splits. The bars represent the number of keywords that are unique to a single split, shared by exactly two splits, or common to all three splits. C) Rank stability of the most consistent keywords across three data splits. Rows correspond to individual keywords, ordered by overall stability (appearance count and average rank). D) Pairwise Jaccard similarity matrix of the keyword sets selected from three independent data splits.

shortness causes them to match as substrings within many unrelated terms, retrieving an excessively large and noisy set of articles. While this boosts nominal coverage, it substantially increases the volume of documents that must be fetched and screened by the downstream LLM classifier, reducing the overall efficiency of the pipeline.

To circumvent this issue, we proposed a hybrid method that restricts the keyword search to the set of top-frequency keywords (such as top 1000) when applying the greedy algorithm. This restriction prevents the selection of overly granular terms that lack practical utility as database search queries, while preserving the coverage efficiency of the greedy strategy. Additionally, we created three splits to assess the stability of our proposed method in using a keyword set to cover the field. We find that, across all three splits, the top 22 selected keywords achieve 90% coverage of papers, and the top 10 achieve 80% (Table S7). The keyword sets show strong consistency: among the keywords pooled from the top-22 lists of the three splits, 15 are common to all three splits, and 23 appear in at least two splits (Fig. 4B). Furthermore, the rankings of these common keywords remain highly consistent across splits (Fig. 4C, 4D). The 15 common keywords achieve over 85% coverage (Table 2). These results suggest that using a relatively small set of keywords can capture most papers, which is especially useful for literature searches aiming to maximize coverage, even at the cost of some noise.

To assess whether the seed keyword set overfits to MeSH labeling biases, we curated an independent positive corpus of 470 articles from synthetic biology-themed journal issues across 22 journals (Table S8), none of which were selected using MeSH indexing. When applying the first-stage keyword retrieval to this corpus, 442 articles (94.04%) were captured. This high coverage demonstrates that the keyword set, although constructed from MeSH-derived articles, generalizes effectively to synthetic biology literature outside the MeSH system.

3.3. Comparative experiment: two-step pipeline versus keyword-only approach

In the third part of this study, we empirically validated the two step pipeline under conditions that simulate real world deployment, since no pre-existing labeled dataset is available for direct end-to-end evaluation. The validation comprised two complementary experiments.

First, we assessed the pipeline's ability to retrieve true synthetic biology articles by using the complete set of papers published in *ACS Synthetic Biology*, a journal exclusively dedicated to the field; these

Table 2
Coverage of the 15 common keywords across three independent data splits.

Keywords	Split 1 (%)	Split 2 (%)	Split 3 (%)	Average (%)
Synthetic	54.03	55.98	54.34	54.78
Engineering	17.32	16.59	17.85	17.25
Genetic	10.64	10.06	10.01	10.24
Logic	10.53	10.11	9.37	10.00
Protein	7.85	6.95	6.85	7.22
Cells	7.16	7.21	6.90	7.09
Microbial	6.85	7.48	6.74	7.02
Bacteria	6.42	5.74	7.11	6.42
DNA	6.42	7.37	5.06	6.28
Chemical	5.79	5.11	4.48	5.13
RNA	5.48	5.74	6.06	5.76
Plant	4.53	5.16	4.95	4.88
Organism	3.90	4.53	3.90	4.11
Biosynthesis	2.32	2.58	2.16	2.35
Life	2.42	2.74	2.74	2.63
Overall coverage	85.73	86.84	85.83	86.13

Note: N = 1,899 for each split. Percentages are calculated as n/N × 100%.

papers therefore constitute a fully positive set. When querying this dataset with only the single keyword "synthetic biology," merely 27.20% of the articles were captured. The published keyword set of Shapira et al. [24] yielded a modest improvement to 32.83%, underscoring the inherent limitations of conventional keyword based search strategies. In contrast, the first step of our method — the hybrid keyword selection strategy — achieved 98.14% coverage of the journal's literature. The full two step pipeline, which couples this keyword retrieval with a fine tuned LLM classifier, ultimately identified 57.81% of the *ACS Synthetic Biology* articles as relevant, demonstrating the practical effectiveness of our approach (Fig. 5A). A document-type analysis (Table S9) showed that non-research items (Correction, Editorial Material, Letter) comprise only 3.7% of the corpus, confirming that the 42% non-retrieval is driven by the Article category itself rather than by non-research noise.

Second, we constructed a negative test set focused on challenging borderline cases. Specifically, we collected articles annotated with the MeSH terms "Biology" and "Bioengineering," both of which are parent terms of "Synthetic Biology" in the MeSH hierarchy. After removing any articles used for training or testing, as well as those explicitly containing the phrase "synthetic biology," we evaluated the false positive rate of the pipeline on these closely related yet non synthetic biology articles. The two step pipeline achieved a false positive rate of only 0.76%, which is comparable to the rates obtained with the single keyword search (0.26%) and the Shapira et al. [24] keyword set (0.35%) (Fig. 5B). This result indicates that our method effectively discriminates against borderline negative articles while preserving high recall, further confirming the robustness of the pipeline when applied to real world data. Because the negative articles drawn from parent MeSH terms "Biology" and "Bioengineering" may not fully represent the disciplinary diversity encountered in real-world databases, we applied the fine-tuned classifier to 1,336,192 articles from the MeSHup dataset without any domain pre-filtering (after excluding training samples and records tagged with "Synthetic Biology"). The model produced 4,264 false positives, yielding a false positive rate of 0.32%. This result demonstrates that the classifier maintains negligible noise even when confronted with a broad, unfiltered multidisciplinary corpus.

We further explored whether a more aggressive model variant could raise recall on the article set derived from *ACS Synthetic Biology*. By fine tuning the Qwen3 4B model with slightly different hyperparameters (using the same training dataset as in Section 3.1), we obtained a variant that increased the two step recall to 74.88%. However, this improvement came at the cost of a higher false positive rate

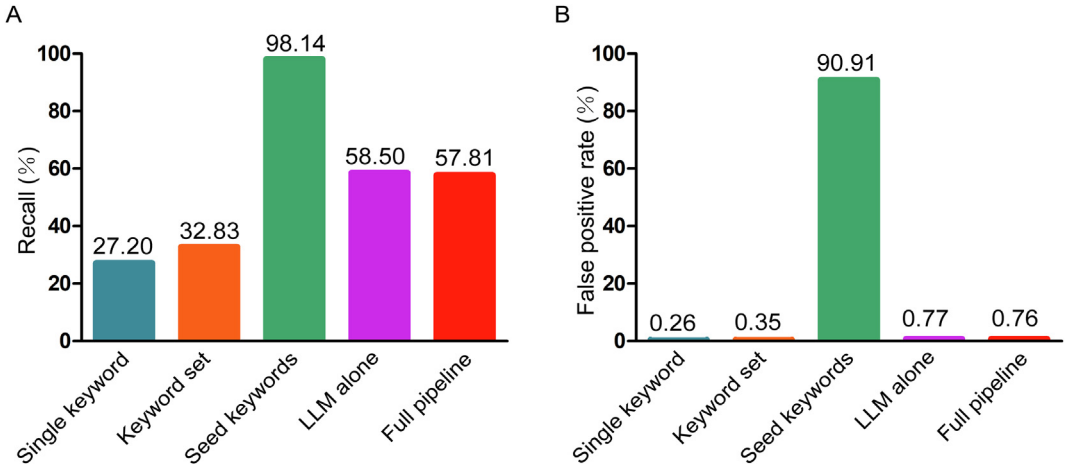


Fig. 5. Performance of different retrieval and classification strategies on two independent validation sets. A) Recall measured on the complete set of articles published in *ACS Synthetic Biology*, which constitutes a fully positive corpus. B) False positive rate evaluated on a curated negative set composed of articles annotated with the parent MeSH terms "Biology" and "Bioengineering," with training, testing, and "synthetic biology" labeled articles excluded. Bars: "Single keyword" (only "synthetic biology"), "Keyword set" (Shapira et al.), "Seed keywords" (this study, step1), "LLM alone" (this study, step2), "Full pipeline" (this study, complete two step pipeline). Abbreviations: *ACS*, American Chemical Society; MeSH, Medical Subject Headings; LLM, large language model.

of 1.59% on the borderline negative corpus (Fig. S1). Although still within an acceptable range, this elevated noise level may introduce unwanted contamination into the curated corpus. We therefore retained the more conservative model (false positive rate 0.76%) for the downstream bibliometric analyses presented in Section 3.4.

To further assess whether the pipeline generalizes beyond the MeSH system, we applied the full two-step pipeline to an independent corpus of 470 articles curated from synthetic biology-themed journal issues across 22 sources (Table S8). This corpus was constructed without using MeSH indexing. The pipeline achieved a recall of 62.97% (296/470) on this external corpus — closely aligned with the 57.81% recall observed on *ACS Synthetic Biology* — suggesting a consistent retrieval ceiling across independent non-MeSH sources.

To determine whether the moderate recall (57.81%) reflects a definitional boundary issue or a methodological failure, we conducted a stratified sampling analysis of the *ACS Synthetic Biology* articles. We defined four decision groups based on the conservative (recall 57.81%, false positive rate 0.76%) and aggressive model (recall 74.88%, false positive rate 1.59%) outputs (Table S10). Group A ($N = 1{,}523$) comprised papers missed by the conservative model; Group B ($N = 2{,}087$) comprised papers captured by it. Groups C and D further partitioned Group A according to the aggressive model's behavior: Group C ($N = 627$) contained papers missed by the conservative model but captured by the aggressive one, while Group D ($N = 896$) contained papers missed by both models. Random sampling ($n = 80$ per group) and manual assessment revealed a clear gradient in boundary-paper proportions: 17.5% in Group B (captured by conservative), 20.0% in Group C (captured only by aggressive), 32.0% in Group A (all missed by conservative), and 41.3% in Group D (missed by both). This pattern indicates that model consensus on negative classification strongly correlates with boundary characteristics. The 1.8-fold enrichment of boundary papers in the negative set (Group A vs. Group B) demonstrates that the conservative model has learned to identify and filter boundary literature. The high boundary proportion in Group D (41.3%) — where both models independently exclude the paper — indicates that the articles most resistant to classification are overwhelmingly boundary or atypical cases rather than misclassified core literature. This suggests that the moderate recall reflects an inherent definitional boundary issue rather than a methodological failure.

3.4. Preliminary landscape mapping of synthetic biology

We applied our two step pipeline to an OpenAlex snapshot, and the resulting curated corpus allowed us to construct a bibliometric landscape of synthetic biology research. Several macro-level patterns emerge from this analysis.

Annual publication trends reveal a consistent and sustained increase in synthetic biology research output over the past two decades, indicating vigorous and accelerating growth in the field (Fig. 6A). The country collaboration network highlights the United States as the dominant contributor, with notably strong co authorship ties with the United Kingdom, along with substantial collaborations involving China, Germany, and other European nations (Fig. 6B).

To uncover the intellectual structure of the field, we performed a co citation network of the top 50 most frequently co cited references, revealing four distinct knowledge clusters: (1) molecular engineering and *in vitro* synthetic biology, (2) DNA nanotechnology and molecular computing, (3) genetic circuit engineering and genome synthesis, and (4) gene editing (Fig. 6C). This data driven structure aligns well with the framework of enabling technologies outlined by Zhang et al. [47]. A keyword co occurrence analysis after removing overly broad terms such as "biology" and "chemistry" exhibits a clear bipartite community structure, as detected by the Walktrap algorithm (Fig. 6D). One cluster centers on molecular-scale construction and characterization, comprising terms such as DNA, peptide, fluorescence, supramolecular chemistry, and nanotechnology, which together represent a bottom-up, *in vitro* paradigm of biomolecular assembly. The other cluster is organized around cellular-scale engineering and genomic programming, featuring synthetic biology, CRISPR, metabolic engineering, *Escherichia coli*, and computer science, reflecting a top down strategy that rewires living chassis organisms. This division captures the field's two foundational approaches. This division mirrors the two foundational technical routes of synthetic biology: bottom up reconstitution from purified components *in vitro* and top down reprogramming of living chassis organisms [48,49]. These concordances between a bibliometric decomposition and an expert–curated classification provide external validation that our curated corpus faithfully reflects the intellectual organization of synthetic biology research.

The thematic map further supports this view. Motor Themes (upper-right, well-developed and central) are dominated by chemistry, biochemistry, materials science, nanotechnology, and DNA, confirming that molecular-level engineering drives the field. Niche Themes (upper-left, well-developed but peripheral) include computer science, engineering, physics, and artificial intelligence, suggesting that computational methods remain specialized but may soon merge into the core. These two themes resonate with several active frontiers in current synthetic biology research, including AI-driven de novo protein design [50], the engineering of miniature CRISPR associated systems for genome editing [51,52], LLM-based knowledge synthesis for biomanufacturing [53,54], and the integration of AI with automated biofoundries [55,56]. Emerging or Declining Themes (lower-left, weakly developed) such as medicine, neuroscience, and pharmacology represent topics that are either nascent and yet unformed or losing prominence, and their status needs to be monitored over time. Basic Themes such as biology serve as transversal foundations that underpin research across all quadrants but offer limited discriminatory power for understanding the field's internal structure (Fig. 6E). Together, the keyword co-occurrence network, co-citation clusters, and thematic map validate the practical utility of our method for mapping research landscapes at scale.

In the course of this landscape analysis, two observations also support the validity of our constructed strategy. First, manual inspection of the 50 most co-cited references revealed that these papers are overwhelmingly concentrated within synthetic biology and its directly allied fields, rather than dispersing across multiple unrelated disciplines (Supplementary data 2). The absence of a fragmented or multi domain clustering pattern in the co-citation network suggests that our method successfully compresses the literature into a coherent knowledge core, rather than diluting the field with peripheral or irrelevant works. Second, an analysis of the most relevant sources shows that *ACS Synthetic Biology*, the flagship journal of the field, ranks second in publication count among all venues (Table S11). The prominence of a dedicated synthetic biology journal in the source list further corroborates the strong thematic concentration achieved by our filtering strategy. Together, these findings indicate that the combination of high-recall keyword selection and high-precision LLM classification effectively captures the intellectual essence of synthetic biology, while minimizing noise from adjacent or unrelated fields.

4. Discussion

The primary contribution of this work is a methodological innovation that integrates the semantic understanding capability of LLMs into a bibliometric pipeline. This aligns with the emerging paradigm in which LLMs serve not merely as text generators but as autonomous agents orchestrating the synthetic biology design-build-test-learn cycle [57]. By providing a semantically curated, high-purity literature corpus, our pipeline addresses the knowledge fragmentation bottleneck that constrains downstream agentic systems — such as automated literature mining, hypothesis generation, and multi-agent orchestration — enabling them to operate on reliable, domain-grounded evidence

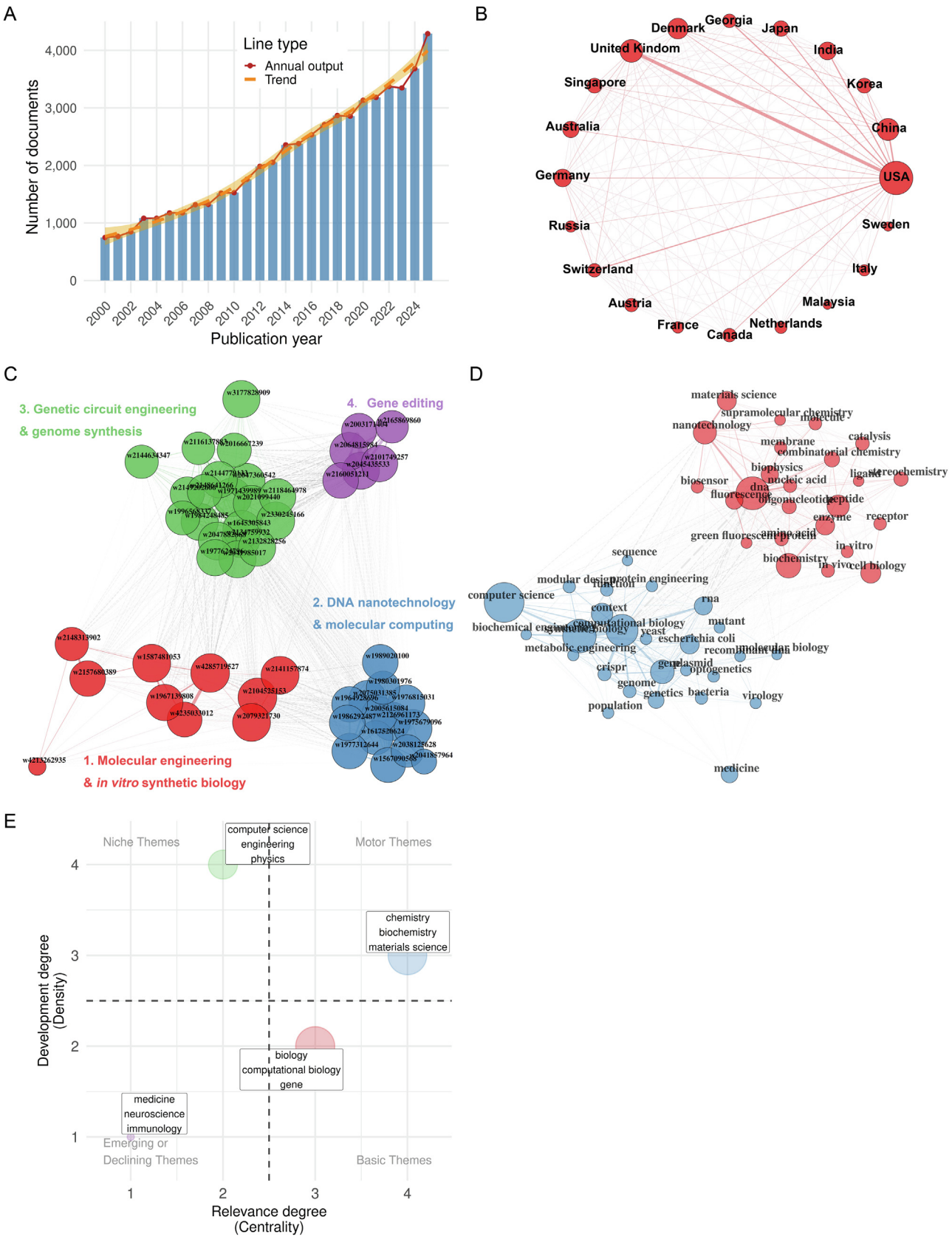


Fig. 6. A) Annual scientific production in synthetic biology (2000–2025). B) Country collaboration network among the top 20 most productive countries. C) Co-citation network of the 50 most frequently co-cited references, revealing four distinct knowledge clusters. D) Keyword co-occurrence network exhibiting a bipartite community structure: one cluster centers on molecular-scale construction, while the other reflects cellular-scale engineering. E) Thematic map identifying central Motor Themes, emerging Niche Themes, peripheral Emerging/Declining Themes, and Basic Themes. All panels were generated using the R package bibliometrix (v5.3).

rather than noisy, keyword-retrieved collections. By pairing a high-recall keyword-based retrieval stage with a fine-tuned LLM classifier, our approach addresses a long-standing challenge in mapping interdisciplinary fields: the inherent ambiguity of domain boundaries. Conventional keyword-based retrieval operates on the assumption that a field can be adequately captured by a predefined set of lexical markers. As we have shown, this assumption systematically fails for synthetic biology, where many technically relevant articles do not carry the expected domain specific keywords. The LLM classifier overcomes this limitation by evaluating the full semantic content of each candidate article rather than relying on the presence or absence of any particular term. It is important to acknowledge that the 0.97 macro-F1 achieved on the curated test set primarily indicates the model's capacity to reproduce MeSH annotations; external validation on independent corpora (see Section 3.3) is necessary to confirm generalization beyond the MeSH system.

Beyond the integration of LLM-based classification, our pipeline introduces a quantitatively validated approach to constructing seed keyword sets — a task that has traditionally been treated as an ad hoc, expert-dependent step in bibliometric studies. Our hybrid greedy-frequency strategy replaces this informal process with an explicit optimization framework and, crucially, provides stability evidence rarely seen in prior work: across three independent data splits, the selected keyword sets showed considerable consistency, with substantial overlap among the top-ranked keywords. This quantitative validation transforms keyword selection from a subjective design choice into a reproducible methodological component, thereby strengthening the overall rigor of the pipeline.

Finally, the proposed framework is inherently transferable to other emerging or ill-defined interdisciplinary fields such as proteomics, brain science, immunoengineering, and nanobiotechnology, all of which share the same fundamental challenge of blurred boundaries and non standardized terminology. Because the keyword selection algorithm operates solely on MeSH terms and author-supplied keywords, and the LLM classifier can be fine tuned on any custom-labeled dataset, the pipeline can be adapted to new domains with minimal modification. We therefore envision this approach not merely as a one time solution for synthetic biology, but as a methodological template that can be instantiated across a spectrum of emerging biological disciplines, enabling more consistent, comprehensive, and semantically grounded bibliometric analyses in fields that resist simple keyword based definition.

It is worth noting that the 57.81% recall of the *ACS Synthetic Biology* corpus — although substantially higher than the single keyword search (27%) and the keyword set of Shapira et al. [24] (33%) — still leaves a considerable fraction of articles unretrieved. This gap can be understood from two perspectives. On the data side, a specialist journal such as *ACS Synthetic Biology* inevitably publishes borderline articles whose classification is inherently subjective; even human experts may disagree on whether certain studies qualify as synthetic biology. Although our training set deliberately included such borderline samples, the ground truth labels themselves reflect judgement calls that are difficult to standardize, placing an upper bound on the achievable recall under any conservative classification strategy.

On the model side, several aspects of our pipeline can be further improved. First, the composition of the training dataset, while deliberately designed to include positive, borderline, and background samples, remains a simplified approximation of the true semantic landscape encountered in real-world literature databases. Because the training data mainly draws upon PubMed-indexed records, in the bibliometric stage we restricted the first-stage keyword retrieval to papers indexed in PubMed, prioritizing reliable performance within a well-matched input space over the riskier, broader coverage that an unrestricted search might provide. Future work could incorporate training data from a wider range of disciplines and develop more robust evaluation protocols to systematically assess the classifier's generalisability, thereby enabling safe deployment across multiple databases. Second, the semantic space of background articles in the real world is vast and heterogeneous, spanning fields as diverse as medicine, agriculture, environmental science, and industrial biotechnology. Our negative training set, though carefully curated, inevitably represents only a fraction of this diversity. Consequently, the LLM classifier may encounter previously unseen semantic patterns during large-scale deployment, potentially leading to misclassifications on out-of-distribution samples. A natural remedy is to expand the diversity of negative instances or to adopt active learning strategies that iteratively introduce the most informative borderline cases. Third, the current classifier is built upon Qwen3-4B, a general-purpose foundation model that has not been specifically pre-trained on scientific or biomedical corpora. Replacing this backbone with a domain-adapted model — such as SciBERT [58], BioBERT [59], or a version of Qwen further pre-trained on scientific literature — may yield additional improvements in both classification accuracy and robustness, particularly for articles that rely heavily on technical jargon or nuanced experimental descriptions. These two directions — more comprehensive training data and stronger domain-adapted models — represent natural extensions of the current pipeline and could further enhance its generalizability across a wider range of emerging interdisciplinary fields.

In summary, the proposed pipeline offers three main advantages: (1) it replaces surface-level keyword matching with meaning-level semantic assessment via a fine-tuned LLM, substantially improving both recall and precision in field delineation; (2) the hybrid greedy-frequency keyword selection strategy provides a quantitatively validated and reproducible alternative to ad-hoc expert-dependent keyword design; and (3) the modular architecture allows straightforward transfer to other interdisciplinary domains. Its current limitations include the 57.81% recall ceiling, which is largely attributable to the inherent ambiguity of borderline cases, the restricted diversity of the training data drawn primarily from PubMed-indexed records, and the use of a general-purpose rather than domain-adapted LLM backbone. Addressing these limitations through more diverse training data and stronger scientific language models represents a natural path forward.

5. Conclusion

In this work, we developed a two-step bibliometric pipeline that combines a hybrid keyword selection algorithm with a fine-tuned LLM to address the long standing challenge of delineating emerging interdisciplinary fields. Taking synthetic biology as a case study, we demonstrated that the pipeline substantially outperforms conventional keyword based methods in both coverage and precision, while producing a curated literature corpus sufficiently robust for large-scale bibliometric landscape analysis. The proposed framework is modular, quantitatively validated, and transferable to other fields that resist simple lexical definition. We therefore envision this approach as a methodological template that can be instantiated across a spectrum of emerging biological disciplines — including those raising biosafety concerns — enabling a more consistent, comprehensive, and semantically grounded understanding of emerging interdisciplinary fields.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

Author contributions

Wenbing Ma: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Pengyuan Nie:** Validation, Software, Data curation. **Rongling Yin:** Validation, Software.

Lijuan Li: Writing – review & editing, Supervision, Conceptualization.
Lei Wang: Writing – review & editing, Supervision, Conceptualization.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bsheal.2026.09.006.

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